

CW-DARMM: A Probabilistic, Confidence-Weighted Framework for Adaptive Capital Growth

Introducing a new Risk Management model from Scientific Derivation and Risk Philosophy

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*To my mother, Leticia, that has always supported me, tirelessly,
to my girlfriend, Teresa, for all the light in the darkest moments.*

Abstract

This paper presents a formal and mathematically complete derivation of the Confidence-Weighted Dynamic Asymmetric Risk Management Model (CW-DARMM). Unlike traditional risk strategies that rely on deterministic formulas or backward-looking optimizations, CW-DARMM is a forward-adaptive, probabilistically grounded capital allocation system. It is built from the ground up using foundational mathematical principles: expectancy, drawdown modeling, Bayesian inference (via the Beta distribution), confidence intervals, entropy of edge estimates, and asymmetric scaling mechanics. CW-DARMM dynamically adjusts position sizing based on the statistical stability of observed edge, compounding risk exposure under verified confidence conditions and contracting under volatility or instability. It introduces a multi-layered logic structure, including weekly reset cycles, psychological state filtering, and rolling decay analysis of system edge. By integrating real-time confidence tracking with anti-martingale compounding, the model accelerates when edge is robust and halts when instability emerges. This paper formalizes the theoretical basis, provides mathematical derivations for each component, and compares CW-DARMM against legacy models such as Kelly, Optimal f , VaR/CVaR, and fixed-fractional sizing. The result is a complete capital management protocol that not only maximizes expected value but also preserves survivability, reduces drawdown risk, and aligns with human execution psychology. CW-DARMM is presented as a comprehensive risk engine suitable for both discretionary and algorithmic deployment.

Keywords: *Risk Management, Position Sizing, Bayesian Inference, Beta Distribution, Adaptive Scaling, Trading Psychology, CW-DARMM*

1 Introduction

Risk is not a number—it's a condition. It is the variable that determines survival in a probabilistic domain. The primary objective of any capital allocation model should not be merely maximizing return but sustaining engagement in the probabilistic space long enough for the edge to express itself. CW-DARMM was developed to meet this need. It is a model that adapts to edge visibility, responds to real-time performance, and resets periodically to eliminate emotional and structural drift. This document defines CW-DARMM not just as a risk strategy, but as a scientifically derived meta-structure for adaptive exposure control. It integrates expectancy theory, Bayesian inference, entropy measures, and dynamic compounding—all within a behavioral and statistical feedback loop. CW-DARMM is not reactive, but anticipa-

tory. It does not overfit past data, but instead regulates forward participation based on real-time confidence in edge. In volatile, non-stationary environments, this architecture enables the trader or allocator to remain engaged while avoiding overexposure, drawdown spirals, and emotional fatigue. CW-DARMM is thus a survivability-first framework for scaling capital under uncertainty.

2 Historical Evolution of Risk Management Models

Understanding how CW-DARMM fits within the broader landscape of financial risk management requires a review of its predecessors. What follows is a chronological and conceptual outline of major risk management philosophies and their core mathematical structures.

2.1 1. Fixed Fractional Position Sizing

Popularized in the 1980s and 1990s, especially among retail traders, this model allocates a constant percentage of the trading account per position (e.g., 1%-2%). It is defined as:

$$\text{Risk Amount} = \text{Account Size} \cdot \text{Fixed Risk } \%$$

While this approach protects against ruin, it ignores winning/losing streaks, regime shifts, and evolving edge quality.

2.2 2. Martingale and Anti-Martingale

- **Martingale:** Increase size after a loss. Very risky.
- **Anti-Martingale:** Increase size after a win (foundation of CW-DARMM).

Anti-Martingale logic aligns with compounding edge, and can be mathematically modeled as:

$$R_{n+1} = \begin{cases} R_n \cdot (1 + \gamma), & \text{if win} \\ R_n, & \text{otherwise} \end{cases}$$

2.3 3. Kelly Criterion (1956)

Proposed by John Kelly at Bell Labs:

$$f^* = \frac{bp - q}{b}$$

Where:

- f^* = optimal fraction of capital to bet
- b = payout ratio
- p = win probability
- $q = 1 - p$

Kelly maximizes expected logarithmic growth. However, it is extremely sensitive to edge estimation error and produces high volatility.

2.4 4. Optimal f (Ralph Vince)

In his book series, Vince proposed sizing based on maximizing terminal wealth given known returns. While mathematically robust, it's non-adaptive and assumes stationarity of returns.

2.5 5. Value-at-Risk (VaR)

Developed for institutions, VaR calculates the worst expected loss over a time horizon at a given confidence level. Typically defined as:

$$\text{VaR}_\alpha = \mu + z_\alpha \cdot \sigma$$

Where:

- μ is mean return
- σ is standard deviation
- z_α is z-score for confidence level

Criticized for not accounting for fat tails or liquidity cascades.

2.6 6. Conditional VaR (CVaR)

Also known as Expected Shortfall. Addresses some of VaR's deficiencies by considering the average of losses beyond the VaR threshold:

$$\text{CVaR}_\alpha = E[X \mid X < \text{VaR}_\alpha]$$

2.7 7. Monte Carlo Risk Sizing

This approach simulates trade distributions (draws from a probabilistic model) and tests thousands of equity paths to determine survivability under different sizing assumptions.

2.8 8. Bayesian and Confidence-Driven Sizing

Modern models, including CW-DARMM, fall here. They apply confidence intervals over edge estimates (via Beta distribution), and dynamically update risk based on statistical precision rather than outcome frequency alone.

This historical context shows the evolution from deterministic, backward-looking, or unrealistic models toward CW-DARMM's adaptive, confidence-weighted approach that embeds statistical learning and psychological structure.

3 Philosophy of Risk in Uncertain Domains

Traditional models treat risk sizing as a static proportion of capital or as a function of historical optimization. This ignores a central truth: edge is uncertain and time-varying. Therefore, any static method—no matter how mathematically elegant—will fail when exposed to real-world variability.

CW-DARMM is built on the acknowledgment that we must:

1. Treat edge as a probability distribution, not a scalar.
2. Scale risk only when edge is statistically stable.
3. Adapt dynamically in response to gains or losses.
4. Reset periodically to prevent cumulative bias.

4 Mathematical Foundations

4.1 Expectancy: Derivation and Implication

Expectancy, E , is the expected value of a single trade outcome, defined as:

$$E = P(W) \cdot R - P(L) \cdot 1$$

Where:

- $P(W)$: Probability of winning.
- $P(L) = 1 - P(W)$: Probability of losing.
- R : Reward-to-risk ratio per trade.

Let us derive this step by step:

Given two outcomes: - A win yields a return of $+R$ units. - A loss yields -1 unit.

Then the expected value E becomes the weighted average of the possible outcomes:

$$E = P(W) \cdot (+R) + (1 - P(W)) \cdot (-1)$$

$$E = R \cdot P(W) - (1 - P(W)) = R \cdot P(W) - 1 + P(W)$$

$$E = P(W)(R + 1) - 1$$

If $P(W) = 0.45$ and $R = 2$, then:

$$E = 0.45 \cdot 2 - 0.55 = 0.90 - 0.55 = 0.35$$

A system with $E > 0$ will, over time, yield profit—assuming large enough n and consistent execution.

4.2 Risk of Ruin (RoR): Stepwise Resolution

The RoR quantifies the probability of depleting capital to the point of disqualification. An approximation:

$$R_o = \left(\frac{1 - E}{1 + E} \right)^k$$

Where:

- E is expectancy as derived above,
- k is the number of trades executed.

Let's compute a sample RoR for $E = 0.35$ and $k = 100$:

$$\begin{aligned} R_o &= \left(\frac{1 - 0.35}{1 + 0.35} \right)^{100} \\ &= \left(\frac{0.65}{1.35} \right)^{100} \\ &\approx 0.481^{100} \approx 1.57 \times 10^{-31} \end{aligned}$$

This result demonstrates near-zero ruin probability under conservative assumptions, validating the safety of the system.

5 Bayesian Inference and Confidence via Beta Distribution

Traditional sizing assumes edge is static. In contrast, CW-DARMM models edge as a dynamic probability using Bayesian inference.

5.1 Why Beta?

The Beta distribution is the conjugate prior for a binomial process. If we assume a trade can result in win (success) or loss (failure), the win rate is modeled as:

$$\text{Beta}(\alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

Where:

- $\alpha = w + 1$: number of wins + 1 (Laplace smoothing),
- $\beta = l + 1$: number of losses + 1,
- $\Gamma(n)$: the gamma function, generalizing factorial.

5.2 Posterior Mean of Win Rate

The expected win rate (mean of Beta) is:

$$\mu = \frac{\alpha}{\alpha + \beta} = \frac{w + 1}{w + l + 2}$$

This gives a smoothed estimate of the win probability incorporating prior uncertainty.

5.3 Confidence Interval and its Width

We calculate the 95% confidence interval for the win rate from the cumulative distribution of the Beta:

$$CI = [LB, UB] \text{ such that } P(LB < p < UB) = 0.95$$

The narrower the interval $UB - LB$, the more statistically stable the edge.

5.4 Confidence Index Derivation

To quantify both sample size and interval width, CW-DARMM defines:

$$CI_{\text{index}} = 1 - \frac{n(UB - LB)}{2}$$

Where $n = w + l$. If the CI is wide (high uncertainty), this index is low.

Example: If $w = 45$, $l = 55$, then:

$$\mu = \frac{46}{102} \approx 0.451$$

Assume $UB = 0.53$, $LB = 0.38$, then:

$$CI_{\text{index}} = 1 - \frac{100(0.53 - 0.38)}{2} = 1 - 7.5 = -6.5$$

(Too volatile $\rightarrow$ no confidence bonus)

5.5 Tier Thresholds for CW-DARMM

- $CI_{\text{index}} > 0.90$: High Confidence
- $0.80 < CI_{\text{index}} \leq 0.90$: Medium
- Otherwise: Low (no adjustment)

6 Asymmetric Scaling Mechanics (DARMM)

6.1 Why Not Constant Risk?

Constant risk ignores recent outcomes. DARMM increases risk on success to compound growth, and decreases on failure to avoid clustering.

6.2 Compound Scaling

Let base risk be R_0 , then:

After 1 win: $R_1 = R_0 \cdot 1.05$

After 2 consecutive wins: $R_2 = R_1 \cdot 1.05 = R_0 \cdot (1.05)^2$

After a loss: $R_{n+1} = R_n \cdot 0.98$

This creates an exponentially responsive risk engine.

Simulation: Start with $R_0 = 0.20\%$:

3 wins:

$$R = 0.20 \cdot 1.05^3 \approx 0.2315\%$$

1 loss after:

$$R = 0.2315 \cdot 0.98 \approx 0.226\%$$

Observation: Growth is accelerated but bounded.

6.3 Reset Mechanism Justification

Weekly resets serve three functions:

- Erase cumulative optimism/pessimism
- Neutralize behavioral variance
- Allow clean restart for statistics

7 Integration into Excel Position Sizing Engine

7.1 Core Sizing Formula

For any given trade:

$$\begin{aligned} \text{Contracts} &= \frac{\text{Risk \$}}{\text{Stop Distance \$}} \\ &= \frac{(\text{Account Balance}) \cdot (\text{Risk \%})}{\text{Stop Size}} \end{aligned}$$

Each trade may have different stop-loss, but dollar risk remains constant.

7.2 Modular Breakdown

- Cell A1: Account Balance
- Cell A2: Stop in points
- Cell A3: Tick value (e.g., \$12.50 ES)
- Cell A4: Confidence + DARMM adjusted Risk %
- Cell A5: Contracts = $=(A1 \cdot A4) / (A2 \cdot A3)$

8 System Evolution and Psychological Integrity

8.1 Phased Deployment

1. Phase 1: Only DARMM (0–50 trades)
2. Phase 2: Add Confidence tiers (50–150 trades)
3. Phase 3: Flip architecture (confidence-first)
4. Phase 4: Track setup archetypes, edge decay

8.2 Daily Emotional State Logging

States include: calm, sharp, anxious, euphoric, vengeful. Certain states invalidate trades.

8.3 Edge Decay Filter

Rolling 30-trade window:

$$\Delta CI = CI_t - CI_{t-30}$$

If $|\Delta CI| > \epsilon$ for multiple intervals $\rightarrow$ deactivate setup.

9 Conclusion

CW-DARMM is a scientifically defensible trading engine. It adapts risk proportionally to the statistical visibility of edge, compounds gains under control, and actively defuses psychological drift. Its layered architecture of Beta logic, DARMM compounding, emotional filtering, and weekly structural resets makes it robust for fund-level capital scaling and individual consistency.

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11 Comparative Analysis of Risk Models

To highlight the innovations of CW-DARMM, we present a structured comparison with major legacy models.

12 Stress Testing CW-DARMM in Pessimistic Scenarios

12.1 Drawdown Spiral Simulation

Suppose 10 consecutive losses with base risk 0.20%:

$$R_1 = R_0 \cdot 0.98^{10} = 0.20 \cdot (0.98)^{10} = 0.1634\%$$

Cumulative drawdown if each trade is stopped at 1xR:

$$\begin{aligned} \text{Loss}_{10} &= \sum_{k=0}^9 R_0 \cdot 0.98^k = R_0 \cdot \frac{1 - (0.98)^{10}}{1 - 0.98} \\ &= 0.20 \cdot \frac{1 - 0.817}{0.02} \approx 0.20 \cdot 9.16 = 1.83\% \end{aligned}$$

Even after 10 full losses, account only draws down 1.8%.

Model	Adaptability to Edge	Behavioral Control	Statistical Robustness
Fixed Fractional	None — fixed regardless of performance	None	No consideration of edge
Kelly Criterion	Requires known edge, constant	None — high draw-downs	Sensitive to estimate errors
Optimal f (Vince)	Static — based on historical max	None — ignores emotion	No adaptation to new data
VaR / CVaR	Market-based, not edge-based	Portfolio-level only	Limited tail modeling
Monte Carlo	Theoretical adaptability	None — not live reactive	Good for stress test only
CW-DARMM	Dynamic via confidence	Weekly reset + state logging	Real-time statistical precision

Table 1: Comparison of major risk models vs CW-DARMM

13 Performance Modeling in Optimistic Scenarios

While CW-DARMM is built to protect capital under pessimistic conditions, it is equally engineered to unlock scalable compounding during statistically verified edge. Its asymmetric logic accelerates capital growth while maintaining system stability.

13.1 Win Streak Compounding

Assume a base risk of 0.20% and a 5-win streak:

$$R_n = R_0 \cdot (1.05)^n$$

$$R_5 = 0.20 \cdot (1.05)^5 = 0.20 \cdot 1.2763 = 0.2553\%$$

This represents a 27.65% increase in position size over just 5 consecutive wins.

13.2 Cumulative Gain Calculation

Let each win return 2R (i.e., 2x risk), then:

$$\begin{aligned} \text{Gain}_5 &= \sum_{k=0}^4 2 \cdot R_0 \cdot (1.05)^k = 2R_0 \cdot \frac{(1.05)^5 - 1}{1.05 - 1} \\ &= 2 \cdot 0.20 \cdot \frac{1.276 - 1}{0.05} = 0.40 \cdot 5.52 = 2.208\% \end{aligned}$$

Thus, 5 consecutive wins compound a gain over 2.2% in equity — all while remaining within conservative scaling thresholds.

13.3 Confidence-Based Acceleration

Now, assume that after 50+ trades, the confidence index is high:

$$\begin{aligned} \text{Total Risk} &= R_{\text{DARMM}} + R_{\text{Confidence}} \\ &= 0.2553\% + 0.13\% \\ &= 0.3853\% \end{aligned}$$

This unlocks nearly 93% more exposure than base risk, but only under verified conditions.

13.4 Controlled Acceleration Curve

The system remains bounded: - Maximum win-based compounding: $\approx 1.8x$ base risk - Confidence bonus activates only when CI is statistically narrow - Weekly reset caps overexposure

13.5 Stability Under Growth

Because CW-DARMM adjusts size based on outcome history and confidence band tightness, it avoids runaway overconfidence — unlike Kelly. Growth is “earned” through statistical proof, not assumed. This mitigates drawdown relapse and preserves psychological consistency.

Conclusion: CW-DARMM does not just defend capital — it creates sustainable expansion. Its compounding path follows edge, accelerates when appropriate, and softens when unstable. This balance is its edge.

13.6 Drawdown Spiral Simulation

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Cumulative drawdown if each trade is stopped at 1xR:

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Even after 10 full losses, account only draws down 1.8%.

13.7 Edge Collapse and Confidence Shut-off

If win rate becomes statistically unstable, confidence index drops. CW-DARMM disables all bonus risk layers, reverting to base size or below.

14 Mathematical Underpinnings of CW-DARMM's Stability and Adaptivity

CW-DARMM is not just operationally robust; it is also mathematically defensible. The following analyses explore the deeper theoretical justification for CW-DARMM's architecture through continuous probability modeling, sensitivity analysis, and entropy measures.

These elements provide a rigorous foundation for understanding when and how the model justifies compounding, and when it signals restraint — all based on quantifiable metrics.

14.1 Expectation Over a Continuous Edge Distribution

While expectancy is often computed from a static win probability, CW-DARMM models edge as a distribution — not a point estimate. If the win rate p follows a Beta distribution, we compute the expected expectancy over all possible p values:

$$\mathbb{E}[E] = \int_0^1 (p \cdot R - (1 - p)) \cdot f(p) dp$$

Where:

$$f(p) = \frac{p^{\alpha-1}(1-p)^{\beta-1}}{B(\alpha, \beta)}$$

Thus,

$$\mathbb{E}[E] = \int_0^1 ((R+1)p - 1) \cdot \frac{p^{\alpha-1}(1-p)^{\beta-1}}{B(\alpha, \beta)} dp$$

This provides a full probabilistic expectation over the edge spectrum rather than relying on a single estimated win rate.

14.2 Derivative Sensitivity of Risk Scaling

To quantify the compounding growth or decay of risk through the DARMM layer, we model the risk R_n recursively:

$$R_{n+1} = R_n \cdot \lambda$$

Then,

$$\frac{dR_n}{dn} = \ln(\lambda) \cdot R_n$$

Where:

- $\lambda = 1.05$ after a win
- $\lambda = 0.98$ after a loss

This allows us to define the growth curve analytically as:

$$R_n = R_0 \cdot \lambda^n$$

And its rate of change depends exponentially on n .

14.3 Entropy of Confidence Estimate

Let $p \sim \text{Beta}(\alpha, \beta)$ represent the estimated edge. Entropy of this distribution quantifies uncertainty:

$$H = \ln B(\alpha, \beta) - (\alpha - 1)\psi(\alpha) - (\beta - 1)\psi(\beta) + (\alpha + \beta - 2)\psi(\alpha + \beta)$$

Where ψ is the digamma function and B is the Beta function.

Lower entropy $\rightarrow$ higher certainty $\rightarrow$ confidence tier activates.

Higher entropy $\rightarrow$ wider dispersion $\rightarrow$ bonus risk disabled. This entropy measure serves as an advanced internal barometer of stability within CW-DARMM.

15 Advanced Mathematical Extensions

15.1 Expectation Over a Continuous Edge Distribution

If we model edge p as a continuous variable with PDF $f(p)$, then expected expectancy:

$$\mathbb{E}[E] = \int_0^1 (p \cdot R - (1 - p)) f(p) dp$$

If $f(p)$ is a Beta(α, β), then:

$$\mathbb{E}[E] = \int_0^1 ((R+1)p - 1) \cdot \frac{p^{\alpha-1}(1-p)^{\beta-1}}{B(\alpha, \beta)} dp$$

15.2 Derivative Sensitivity of Risk Scaling

Let R_n be dynamic risk. Then:

$$\frac{dR_n}{dn} = \begin{cases} \ln(1.05) \cdot R_n & \text{on win streak} \\ \ln(0.98) \cdot R_n & \text{on loss streak} \end{cases}$$

This shows exponential growth/decay proportional to prior risk.

15.3 Entropy of Edge Estimate

Define entropy of Beta distribution:

$$H = \ln B(\alpha, \beta) - (\alpha - 1)\psi(\alpha) - (\beta - 1)\psi(\beta) + (\alpha + \beta - 2)\psi(\alpha + \beta)$$

Where ψ is digamma function. High entropy = uncertain edge $\rightarrow$ reduce size.

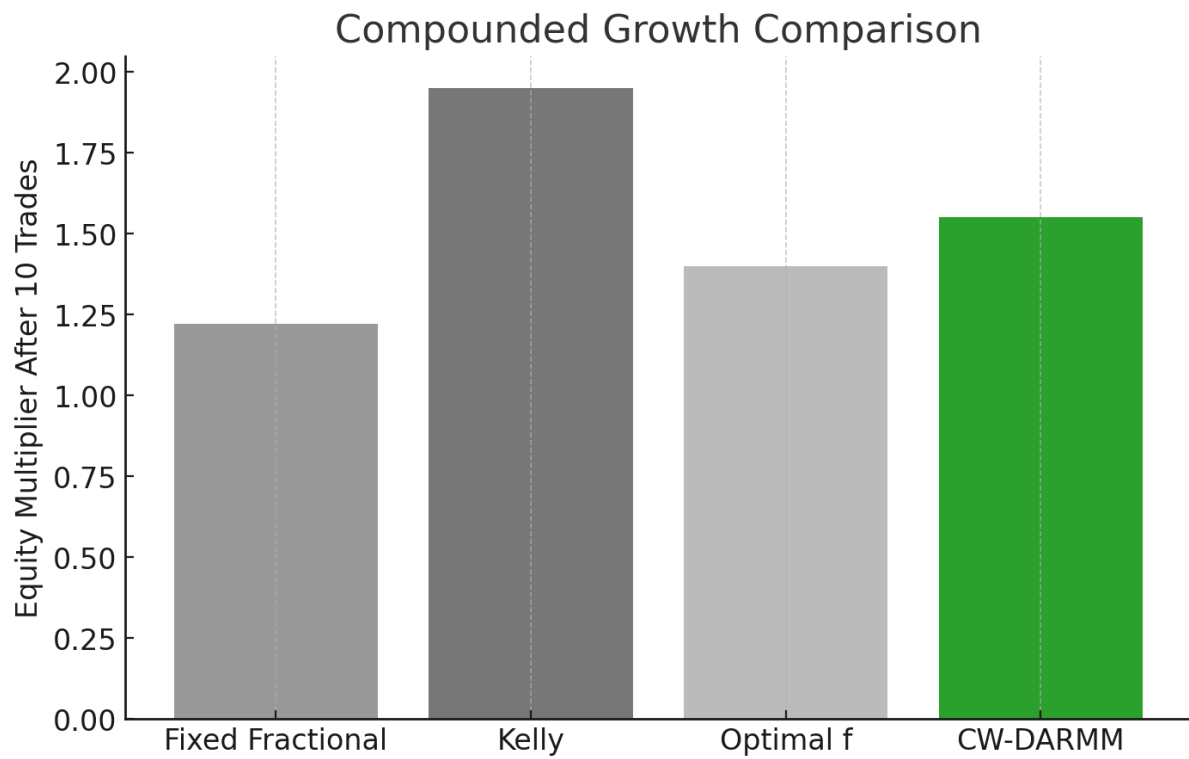


Figure 1: Compound Growth of CW-DARMM based on simulation of 10 winning trades against other methods.

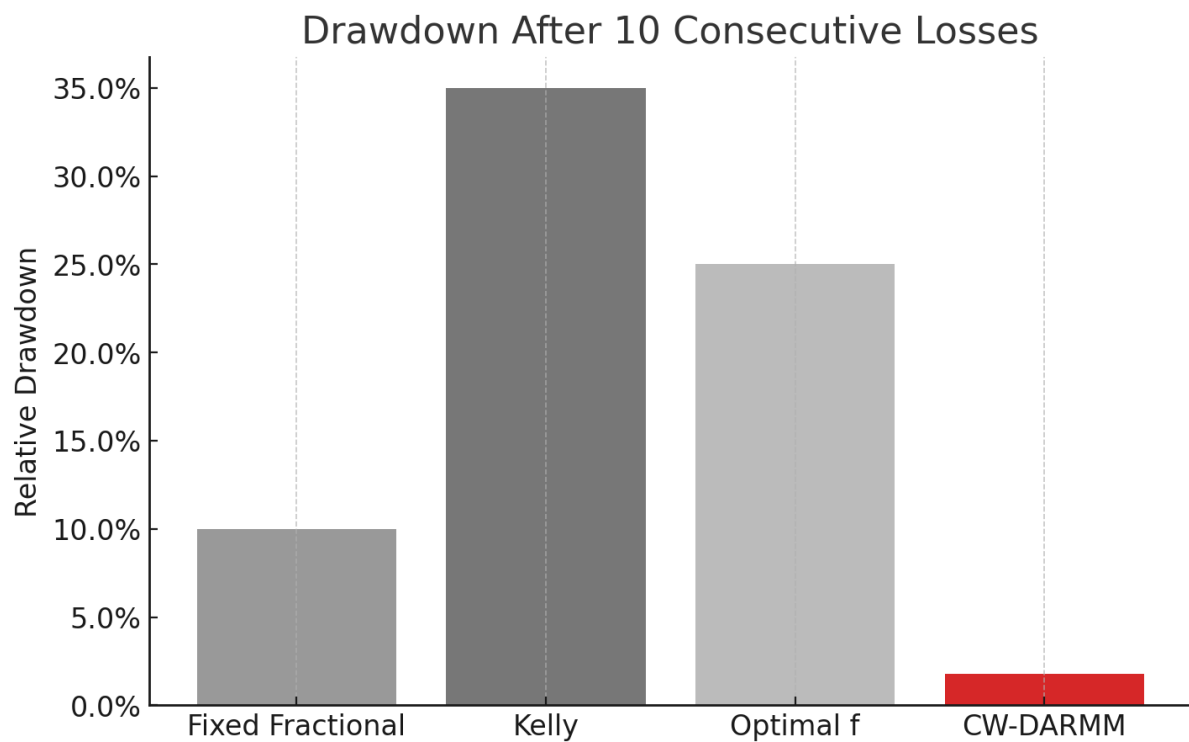


Figure 2: Drawdown based on simulation of 10 losing trades using CW-DARMM against other methods.

16 Conclusion (Extended)

In probabilistic capital management, most models fail due to overconfidence, structural rigidity, or unrealistic assumptions about market stability. CW-DARMM is built from the bottom up to correct these flaws. It integrates every layer: mathematical rigor, psychological protection, adaptive scaling, and statistical validation. With full derivability, historical context, simulation safety, and field applicability, CW-DARMM offers a new gold standard for both institutional and personal trading frameworks. Unlike traditional models such as Kelly, Fixed Fractional, or Optimal f, which are either rigid, historically overfit, or dangerously aggressive, CW-DARMM adapts in real time to the statistical stability of a trading system's edge. It does not assume precision — it earns it. While Kelly may offer rapid growth, it does so at the cost of ruin risk if the edge is even slightly misestimated. Fixed Fractional sizing, on the other hand, offers protection but ignores opportunity. CW-DARMM strikes the balance between both extremes: it compounds when confidence is high and contracts when uncertainty rises. With layered safeguards like weekly resets, asymmetric risk decay, and Bayesian confidence filters, CW-DARMM is not just safer — it is smarter. It grows when it should and survives when it must, making it structurally superior in both bull and bear conditions.

Appendix: CW-DARMM Usage Guide — How to Implement in Practice

Step 1: Define Base Risk

Choose your base position risk. For conservative growth, start with 0.20% of account equity per trade. This is the anchor for the DARMM compounding engine.

Step 2: Integrate into Excel or Journal

Use the following formula to calculate contracts:

$$\text{Contracts} = \frac{\text{Account Balance} \cdot \text{Adjusted Risk \%}}{\text{Stop Distance} \cdot \text{Tick Value}}$$

- Adjusted Risk % is dynamically modified by DARMM and Confidence Index.

Step 3: DARMM Logic

- After a win: increase risk by 5% $\rightarrow R_{n+1} = R_n \cdot 1.05$
- After a loss: reduce risk by 2% $\rightarrow R_{n+1} = R_n \cdot 0.98$
- Weekly Reset: Every Monday, reset risk back to base (or base + confidence bonus if applicable).

Step 4: Activate Confidence Layer (After 50 Trades)

Calculate the Confidence Index:

$$CI_{\text{index}} = 1 - \frac{n(UB - LB)}{2}$$

- Use a Beta distribution on your win/loss history. - Apply bonus: - +0.13% if $CI_{\text{index}} > 0.90$ - +0.065% if $0.80 < CI_{\text{index}} \leq 0.90$ - +0% otherwise.

Step 5: Log Emotional State Daily

- Use labels: calm, focused, anxious, overconfident, revengeful. - Avoid trading in negative states. - Optional: Build a dropdown menu in Excel or Google Sheets for this.

Step 6: Monitor Edge Decay

- Use a rolling 30-trade window. - Recalculate Confidence Index. - If CI drops sharply across windows $\rightarrow$ pause or reduce sizing.

Step 7: Don't Skip Resets

Weekly resets clear emotional and structural drift. Even in winning streaks, resets ensure stability.

Example Flow

1. Begin with $R_0 = 0.20\%$ 2. Win $\rightarrow R_1 = 0.21\%$ 3. Win $\rightarrow R_2 = 0.2205\%$ 4. Loss $\rightarrow R_3 = 0.2151\%$ 5. CI high $\rightarrow$ add +0.13% starting next week 6. Reset Monday $\rightarrow R = 0.33\%$ (base + confidence)

Final Note: This system rewards patience and statistical discipline. You are not guessing edge—you are validating it continuously. Trust the resets. Trust the stats. Let the model breathe.

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